Fiscal Year:	FY 2009	**Task Last Updated:**	FY 06/19/2009
PI Name:	Lee, Yong Jin Ph.D.		
Project Title:	Wearable Beat to Beat Blood Pressure Monitor		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94303-3638	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	02/25/2009	**End Date:**	02/25/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX08CC15P		
Performance Goal No.:			
Performance Goal Text:			
	A key component of NASA's human exploration programs is a system that monitors the health of the crew during space missions. The wearable beat-to-beat blood pressure monitor proposed by Linea Research Corporation can be used to continuously monitor the physiological effect of prolonged space missions (including exposure to reduced gravitational environments) and the effectiveness of its countermeasures. During Phase I of the program, we demonstrated the feasibility of a novel, non-invasive beat-to-beat blood pressure monitor using a lightweight (<40 g) and low-power (<100mA @ 5V) initial prototype. Measurement of beat-to-beat blood pressure was successfully verified during human studies (tilt table tests) using a cuff-based blood pressure monitor as reference. Measurement of the beat-to-beat blood pressure was also verified against an arterial line in animal (porcine) studies using epinephrine to induce blood pressure changes. During Phase II of the program, we will develop and fabricate a field-capable, light-weight, wearable beat-to-beat blood pressure monitor that will be ready for demonstration on space missions. In addition to supplying the blood pressure monitors for use by NASA, Linea plans to introduce the technology for use in ambulatory blood pressure		

Task Description:	monitors. Linea will subsequently introduce the technology for integration into high acuity as well as home based blood pressure monitors. One patent has been filed and three are currently being prepared to protect the intellectual property. A business plan has been prepared to address the commercialization opportunities. POTENTIAL NASA COMMERCIAL APPLICATIONS: Autonomous medical care for the crew during human exploration missions is critical in preventing degradation in health due to adverse physiological responses to space flight environments. A wearable physiological monitor that provides continuous blood pressure (and heart rate) measurements will be extremely valuable in providing proper medical support for both normal activities and medical emergencies. The device can be used to monitor the long-term physiological effects of hypogravity and the effects of the countermeasures against hypogravitational environments. The device can also be modified to monitor the health status of crew during extravehicular activities. The device will provide wireless connectivity to a monitor (e.g. laptop already onboard) to allow transfer of both real-time and historical physiological data. The system will be developed as a single-unit wearable device which necessitates low power consumption and compact and lightweight form factor. The size criterion is a key factor for space applications as launch costs are directly determined by the weight of the device.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed device has the potential to fundamentally change the half-billion dollar blood pressure monitoring device market. The technologies developed under the NASA program are most immediately applicable to ambulatory blood pressure monitors, but can also be applied to high acuity (e.g. arterial line replacement) as well as home blood pressure monitoring devices. For the ambulatory and home blood pressure monitoring segment, the technology developed under this program will enable beat-to-beat blood pressure measurement which will allow continuous measurement of blood pressure. Another significant advantage of the proposed beat-to-beat technology is that it does not require the inflation of the cuff except for calibration measurements. The inflation of the cuff can be uncomfortable for many users and can result in complications such as petechiae on the arm. The key advantage of the proposed technology over arterial lines used for high-acuity applications is the non-invasive nature of the measurement. Blood pressure monitoring based on arterial lines involves invasive procedures requiring percutaneous insertion of catheters into the radial or brachial arteries. Cannulation of the artery requires a skilled health professional and can often take significant time. In addition, cannulation can result in a number of complications including bleeding, infection, and rarely but significantly, lack of blood flow to the tissue supplied by the artery.
Task Progress:	New project for FY2009. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)